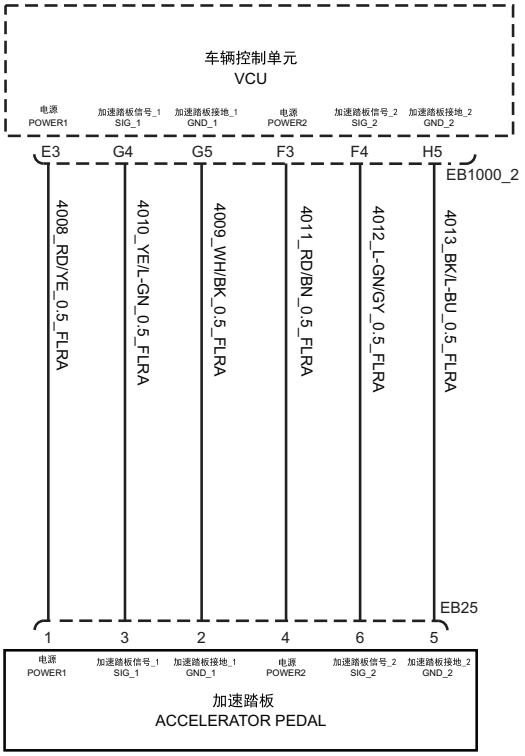
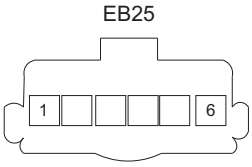
P2123, P2122、 P2128、 P2127、 P2135、 P2136、 P2138

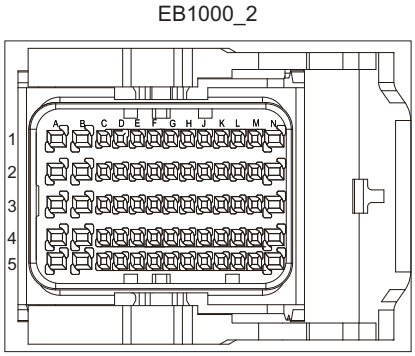
Fault diagnosis code
P2123：DEM_stApdAccrPedlSen1SBCfmd
P2122：DEM_stApdAccrPedlSen1SGCfmd
P2128：DEM_stApdAccrPedlSen2SBCfmd
P2127：DEM_stApdAccrPedlSen2SGCfmd
P2135：DEM_stApdAccrPedlSen1Orng
P2136：DEM_stApdAccrPedlSen2Orng
P2138：DEM_stApdAccrPedlPlaus
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • VCU module fault • Accelerator pedal fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P2123
Accelerator pedal position sensor voltage is greater than or equal to 4.9V
P2122
Accelerator pedal position sensor 1 voltage is lower than or equal to 0.1V
P2128
Accelerator pedal position sensor voltage is greater than or equal to 4.9V
P2127
Accelerator pedal position sensor 2 voltage is lower than or equal to 0.1V
P2135
Any of the following conditions are satisfied: 1.Accelerator pedal position sensor 1 voltage is greater than 4.5V； 2.Accelerator pedal position sensor 1 voltage is lower than 0.5V；
P2136
Any of the following conditions are satisfied: 1.Accelerator pedal position sensor 2 voltage is greater than 4.5V； 2.Accelerator pedal position sensor 2 voltage is lower than 0.5V；
P2138

Any of the following conditions are satisfied: The ratio of“Accelerator pedal signal1/accelerator pedal signal2”>2.2; The ratio of“Accelerator pedal signal1/accelerator pedal signal2”< 1.8;
To set the effect of a fault code condition
Description of circuit diagram
The VCU module monitors the accelerator pedal via low level reference circuit and signal circuit.

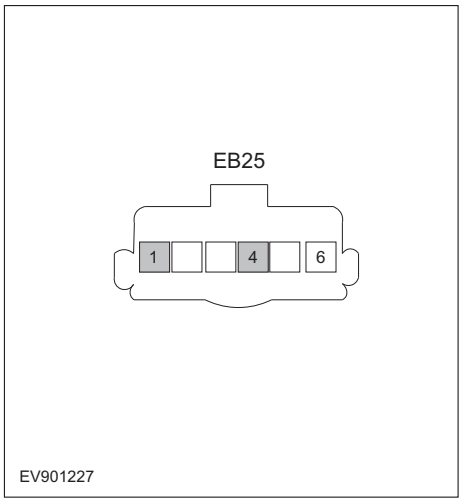
Circuit diagram

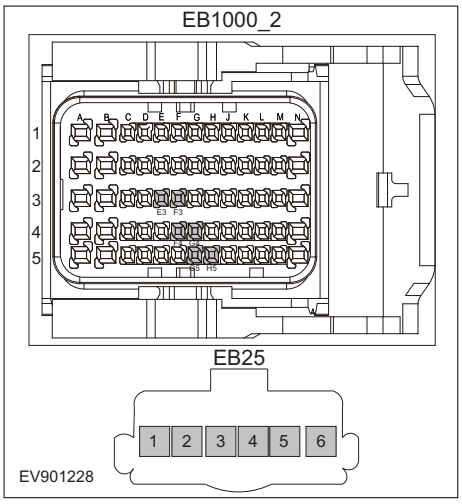


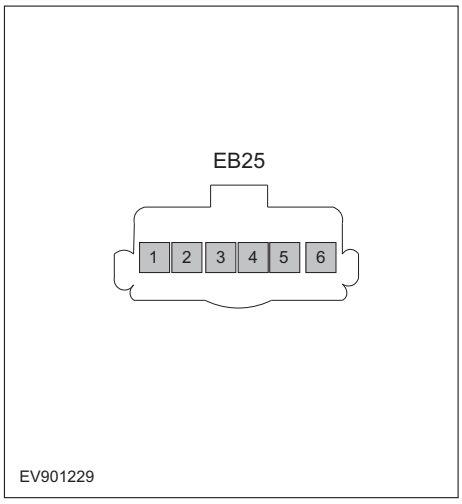


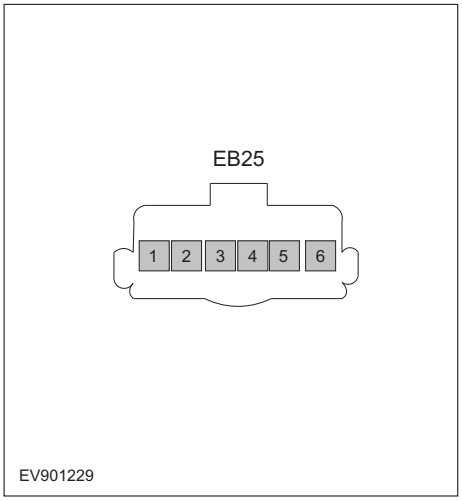


EV901205

<table>
<tr><td colspan="3">Troubleshooting steps</td></tr>
<tr><td>Step 1</td><td colspan="2">General inspection.</td></tr>
<tr><td colspan="2"></td><td>1 Check the fuse for damage.
2 Check battery capacity.
3 Check the harness plug of accelerator pedal, VCU for damage, poor contact, aging and looseness.
4 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO ➞ Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
<tr><td>Step 2</td><td colspan="2">Check fault codes.</td></tr>
<tr><td colspan="2"></td><td>1 Connect the fault diagnosis instrument to the diagnosis interface.
2 Turn the ignition switch to ON position.
3 Diagnose the VCU with diagnostic apparatus.
4 If there is any fault code except for P2123, P2122、P2128、P2127、P2135、P2136、P2138?</td></tr>
<tr><td colspan="2"></td><td>YES ➞ Refer to:DTC Summary list(VCU).</td></tr>
<tr><td colspan="3">NO ↓</td></tr>
<tr><td>Step 3</td><td colspan="2">Check the supply power of brake vacuum sensor when it is started or operating.</td></tr>
<tr><td colspan="2"> 
</td><td>1 Turn the ignition switch to OFF.
2 Disconnect the harness plug EB25 of brake vacuum sensor.
3 Turn the ignition switch to ON.
4 Measure the voltage between the plug terminal of brake vacuum sensor and ground.
Measuring circuit: voltage between terminal 1 on plug EB25 and ground.
Measuring circuit: voltage between terminal 4 on plug EB25 and ground.
Standard value:4.7~5.2V
5 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO ➞ Repair the circuit.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
</table>

Step 4	Check the circuit of accelerator pedal (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1000_2 of VCU module. 3 Measure the resistance between the harness terminal of accelerator pedal and the VCU. Measuring circuit: resistance between terminal 1 on plug EB25 and terminal E3 on plug EB1000_2. Measuring circuit: resistance between terminal 2 on plug EB25 and terminal G5 on plug EB1000_2. Measuring circuit: resistance between terminal 3 on plug EB25 and terminal G4 on plug EB1000_2. Measuring circuit: resistance between terminal 4 on plug EB25 and terminal F3 on plug EB1000_2. Measuring circuit: resistance between terminal 5 on plug EB25 and terminal H5 on plug EB1000_2. Measuring circuit: resistance between terminal 6 on plug EB25 and terminal F4 on plug EB1000_2. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check the circuit of accelerator pedal (short circuit).
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness terminal of accelerator pedal and ground. Measuring circuit: resistance between terminal 1 on plug EB25 and ground. Measuring circuit: resistance between terminal 2 on plug EB25 and ground. Measuring circuit: resistance between terminal 3 on plug EB25 and ground. Measuring circuit: resistance between terminal 4 on plug EB25 and ground. Measuring circuit: resistance between terminal 5 on plug EB25 and ground. Measuring circuit: resistance between terminal 6 on plug EB25 and ground. **Standard value: ∞** 3 Measure the voltage between the harness terminal of accelerator pedal and ground. Measuring circuit: voltage between terminal 1 on plug EB25 and ground. Measuring circuit: voltage between terminal 2 on plug EB25 and ground. Measuring circuit: voltage between terminal 3 on plug EB25 and ground. Measuring circuit: voltage between terminal 4 on plug EB25 and ground. Measuring circuit: voltage between terminal 5 on plug EB25 and ground. Measuring circuit: voltage between terminal 6 on plug EB25 and ground. **Standard value: ≈ 0V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 6	Check the accelerator pedal (components).



1 Measure the resistance of the terminal of accelerator pedal directly.

Measuring circuit: resistance between terminal 2 and terminal 3 on plug EB25.

Measuring circuit: resistance between terminal 5 and terminal 6 on plug EB25.

Standard value:

2 Measure the resistance between the accelerator pedal terminal and the housing.

Measuring circuit: resistance between terminal 1 on plug EB25 and the housing.

Measuring circuit: resistance between terminal 2 on plug EB25 and the housing.

Measuring circuit: resistance between terminal 3 on plug EB25 and the housing.

Measuring circuit: resistance between terminal 4 on plug EB25 and the housing.

Measuring circuit: resistance between terminal 5 on plug EB25 and the housing.

Measuring circuit: resistance between terminal 6 on plug EB25 and the housing.

Standard value: ∞

3 Check whether the result is normal or not?

NO → Replace accelerator pedal.

YES ↓

Step 7	Test whether VCU is norm.

1 Turn the ignition switch to OFF.

2 Disconnect and check the harness plug EB1000_1, IP55_A of VCU.

3 Check and repair.

- Corrosion(install new adapting piece or terminal-cleaning module and pin).
- Pin damaged or bent-install new terminal/pin.
- Fly out of pins-install new pins as needed.

4 Reconnect the VCU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed.

5 Run the system and determine if the problem persists?

YES → Change the VCU Module

NO ↓

At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.
Next Step ↓
Diagnosis end.